

Characterizing the U.S. graduate-level quantum curriculum: Insights into doctoral programs' coursework and comparisons with undergraduate preparation

Alexis Buzzell¹, Timothy J. Atherton², and Ramón Barthelemy¹

¹*Department of Physics & Astronomy, University of Utah, Salt Lake City, Utah, USA*

²*Department of Physics & Astronomy, Tufts University, Medford, Massachusetts, USA*



(Received 6 December 2025; accepted 20 August 2026; published 25 September 2026)

The completion of graduate-level coursework is an important milestone on the path to a doctoral degree in physics. Yet the structure and content of the graduate curricula remain largely undocumented. Here we provide the first national-scale analysis of the graduate-level quantum curriculum, drawing on catalogs from 184 institutions, 121 core quantum course syllabi, and 91 quantum field theory (QFT) course syllabi. Our results reveal the majority of graduate programs require two quantum courses (60.3%) and most commonly use Sakurai's *Modern Quantum Mechanics* textbook (41.2%). Core quantum syllabi converged on four canonical topics: kinematics, angular momentum, approximations, and dynamics. While QFT may be considered advanced coursework that is not necessary for all research fields, it remains essential for describing relativistic and multiparticle systems, yet it is nearly absent from the physics education research (PER) literature. We found that 39% of graduate programs had no formal QFT course offering. Statistical analyses revealed that smaller graduate programs with less research activity were more likely not to offer a QFT course, raising concerns about equity and access. Among institutions that did offer QFT courses, a canonical set of topics emerged: interacting fields, Feynman diagrams, quantum electrodynamics, and radiative corrections. We provide comparisons to the undergraduate quantum curriculum, as well as recommendations to instructors and programs.

DOI: [10.1103/1ww9-cf72](https://doi.org/10.1103/1ww9-cf72)

I. INTRODUCTION

The completion of graduate-level coursework is an important milestone on the path to a doctoral degree in physics. Yet the structure and content of the graduate curricula remain largely undocumented. While undergraduate curricula and instruction have been studied in detail, there has been no comprehensive effort to characterize the graduate curricula, despite its significance in preparing students for careers in both academia and industry. This absence of evidence matters. Graduate coursework can either serve as a gateway to advanced research or as a gatekeeper for persisting in physics. Variability in curricula may reflect healthy diversity based on the research fields of the department, but it may also contribute to uneven preparation. This paper will focus on the characterization of the graduate quantum curriculum, but future studies will characterize other key courses.

In this paper, we present the first national-scale map of the United States (U.S.) graduate quantum curriculum. By analyzing course catalogs and syllabi from 184 doctoral-granting institutions, we identify both shared foundations and differences in graduate training. We find that while core quantum mechanics instruction is broadly standardized, quantum field theory (QFT) varies in availability, raising questions about equity, local research priorities, and the evolving purpose of graduate coursework in physics.

In this paper, we will first review prior literature on graduate physics education, undergraduate quantum curricula, and the limited existing work on QFT instruction. We then describe the methods used to identify doctoral programs of interest for the study, collect course materials, and analyze both core quantum mechanics and QFT curricula across institutions. Next, we present the results of the curricular analysis, including the topics most commonly taught, textbook usage, and relationships between institutional characteristics and course offerings. Finally, we discuss the implications of these findings for graduate physics education, considerations for curricular alignment, and directions for future research.

Published by the American Physical Society under the terms of the [Creative Commons Attribution 4.0 International license](https://creativecommons.org/licenses/by/4.0/). Further distribution of this work must maintain attribution to the author(s) and the published article's title, journal citation, and DOI.

II. LITERATURE REVIEW

A. Graduate physics programs

Research on characterizing graduate physics programs has primarily focused on admission practices, such as the use of the physics Graduate Record Examinations or rubric-based holistic review [1–5]. For example, Bridges *et al.* have been working to characterize the landscape of assessments within U.S. graduate physics programs [6]. Preliminary findings have demonstrated a strong focus of core course requirements on quantum mechanics and electrodynamics, with quantum mechanics required by all programs in the study. The importance of quantum mechanics extends beyond courses, as 90% of U.S. graduate programs test students' quantum knowledge through qualifying exams [7].

B. Undergraduate U.S. quantum mechanics curriculum

Our previous work characterized the modern physics curriculum [8] and the undergraduate quantum mechanics curriculum [9] at U.S. research-intensive institutions using syllabi for topic characterization. The modern physics study found 94% of the 127 research-intensive institutions included quantum concepts in their modern physics courses [8]. With 70% of these courses being required for physics majors to graduate with their four-year degrees, we concluded that the majority of undergraduate physics students enrolled in U.S. research-intensive institutions are first formally introduced to quantum concepts within their modern physics courses.

As a continuation of the modern physics study [8], the undergraduate quantum mechanics study sought to characterize all courses covering quantum concepts required for students to earn a four-year degree at 188 U.S. institutions [9]. This included the modern physics course and upper-level undergraduate quantum mechanics courses. Using course catalogs, we found that all institutions in the study required one course, 92% required two courses, and 50% required three courses.

The undergraduate quantum study utilized syllabi and course catalog data to determine the number of classroom hours spent on quantum concepts within an institution's curriculum required for a four-year degree [9]. It was determined that the average time spent in the classroom on quantum concepts was 63.5 h, but with a standard deviation of 28.1 h. Therefore, it was concluded that the time spent on quantum concepts in the undergraduate curriculum is highly variable across the U.S.

The study also characterized the topics taught in the undergraduate quantum curricula, using syllabi for analysis [9]. The most commonly taught topics included the Schrödinger equation, three-dimensional quantum mechanics, formalisms, and uncertainty principle. One interesting finding was that only 28% of curricula

explicitly taught the Stern-Gerlach experiment (SGE). A study by Zhu and Singh [10] found that students demonstrate a more expertlike understanding of the distinction between physical space and Hilbert space when introduced to the SGE. Dirac notation, an important topic for expertlike understanding of quantum mechanics, was only found in 34% of the institutions' curricula [9]. Despite ongoing efforts to promote a spin-first approach [11,12], the undergraduate quantum curricula study found the traditional position-first approaches were still more common as they were used by 73.7% of instructors [9].

A complementary study by Kruse *et al.* [13] characterized upper-level undergraduate quantum mechanics courses by surveying 76 quantum mechanics instructors. Both Kruse *et al.* [13] and our study [9] found Griffiths [14] to be the most used textbook, followed by McIntyre [15] and then Townsend [16]. They also identified a similar set of topics covered and that position-first approaches were more commonly used [13]. Kruse *et al.* also expanded from our findings. The instructors they surveyed most commonly taught small classes, with 10 or fewer students. They determined prerequisites courses for upper-level quantum mechanics, finding differential equations, mathematical methods, and linear algebra to be the most common. They also found statistically significant differences in topic coverage between spins-first and positions-first courses.

C. Research on interdisciplinary quantum education

While quantum is an integral subject within physics, the development of quantum technologies requires interdisciplinary teams [17]. As a result of this emerging workforce need, interdisciplinary quantum courses are proliferating at universities across the U.S. These courses are listed within institutional catalogs by a variety of names, such as quantum information science and engineering (QISE), quantum technology, science, and engineering, or quantum computing. Similar to our work to landscape the U.S. quantum curricula at all levels in physics departments, Piña *et al.* have been working to landscape the interdisciplinary QISE courses across the U.S. [18]. Utilizing course catalogs from 1456 U.S. institutions, they found 61 institutions offering QISE degree programs, mostly at doctoral-granting schools. Work by Meyer *et al.* has surveyed instructors of QIS courses to determine content covered [19], as well as opportunities and challenges of teaching these courses [17]. They concluded that, as a result of diversity in both instructor and student backgrounds, no canonical introduction to QIS course emerged.

D. Quantum field theory

Our project also sought to understand graduate level QFT courses. While not every student is required to enroll in QFT—as it is a research discipline specific course—it is important to many prominent fields in physics. QFT is

important for condensed matter, high energy physics, cosmology, quantum gravity, and more [20].

The systems found in nature are often multiparticle and/or relativistic systems [20]. QFT is necessary to describe these systems. Making QFT courses an important cornerstone in many physicists' training. Yet, QFT remains an underexamined area in physics education research (PER).

Current PER literature contains only limited references to QFT. For example, Sirnoorkar *et al.* mention QFT and Feynman diagrams only briefly in a discussion of diagram-based reasoning and its importance in physics [21]. Similarly, Verostek *et al.* describe a participant who expressed interest in studying QFT while searching for a graduate research group [22]. In our previous study of the undergraduate curriculum, QFT was identified in only 2% of institutions' required quantum curricula, where it was introduced only at an introductory level [9].

A more in depth PER study on QFT was Lepage's use of active learning strategies in a graduate QFT course [23]. In Lepage's course, students completed readings and online questions before class so that class time could be spent working through problems. Lepage wanted students to practice the decisions and skills similar to those of a theoretical physicist. Using an interactive classroom allowed for students to receive quick instructor feedback. The course coverage was not compromised and Lepage notes active teaching techniques are well suited to other advanced courses.

III. METHODS

A. Institution selection and demographics

Following the methods of our prior work [8,9,24], the *U.S. News* rankings of "The Best Physics Programs" were used as a reference list for institutions in this study [25]. This list comprised 190 institutions that were primarily research-focused and offered graduate-level physics education. However, nine of these institutions offered master's degrees in physics but not doctoral degrees. Although coursework is also important at the master's level, we limited this study to institutions offering doctoral degrees because the next phase of this research examined faculty beliefs about whether the curriculum adequately prepares students for dissertation-level research [26]. Additionally, one institution was a primarily Spanish-speaking institution. Therefore, from the list of 190 U.S. physics programs, this paper analyzes the curriculum of 180 programs.

Our research group also understands the uniqueness of Minority Serving Institutions (MSIs) and wanted to ensure the experiences at MSIs are better captured in our data. To do so, the course catalogs of all U.S. MSIs [27] were referenced to determine which offered doctoral degrees in physics. Of this list, four new institutions were added to our dataset, for a total of 184 institutions analyzed in this paper.

From these 184 institutions, 89% of physics Ph.D. degrees and 53% of exiting physics master's degrees were awarded in the 2023–2024 academic year [28]. These institutions enrolled 84% of all physics Ph.D. students in the U.S. and 79% of first-year physics Ph.D. students during the 2023–2024 academic year. These institutions are primarily public (70%) and have very high research activity (76%) according to the Carnegie Classification of Higher Education Institutions [29]. The other institutions in our dataset include 21% with high-research activity and 3% whose research activity is not defined by these metrics. Furthermore, 18% are MSIs ($n = 31$), including Hispanic Serving Institutions (HSIs, $n = 18$), Asian American and Native American Pacific Islander-Serving Institutions (AANAPISIs, $n = 14$), Historically Black Colleges and Universities (HBCUs, $n = 3$), Predominantly Black Institutions (PBIs, $n = 2$), and Alaska Native-Serving Institutions or Native Hawaiian-Serving Institutions (ANNHs, $n = 2$) [27].

B. Core quantum course data collection and coding

Core quantum courses were defined as courses that covered quantum concepts and that *all* graduate students within a program were required to complete, regardless of research track or specialty. To determine the core quantum courses within each program, the institutions' course catalogs, graduate bulletins, physics department graduate student handbook, physics departmental websites, and/or contact with department administration were utilized to discern graduate-level core quantum courses from graduate-level elective courses.

Using the list of all core quantum courses in the 184 programs, course syllabi were obtained for all core quantum courses required at 68 institutions. The number of core quantum courses required will later be shown in the results to range between 0 and 3 courses; we will therefore analyze 121 syllabi across these 68 institutions. From these 68 institutions, 37% of physics Ph.D. degrees and 21% of exiting master's degrees were awarded in the 2023–2024 academic year [28]. The populations of these programs comprise 32% of first-year physics graduate students and 34% of the total U.S. physics graduate enrollment. These institutions are 16% MSIs, with HSI ($n = 6$), AANAPISI ($n = 4$), AANH ($n = 1$) represented [27]. These 67 institutions are primarily public (74%) and have very high research activity (81%) [29]. The remaining institutions are classified as high-research activity (16%) or are not classified by these metrics (3%).

From these 68 institutions, the first core quantum course syllabus within the core quantum course sequence was analyzed to determine the required textbook for the course. To consolidate the results, unique editions of the textbooks were not disaggregated within the data presented in this paper.

To determine the topics taught, an emergent coding method was used [30]. All topics mentioned in the syllabi were recorded. Using the table of contents from Sakurai [31] (the most commonly utilized textbook, as shown in the results), closely related topics were combined together. Table I below presents the overarching themes within the syllabi, along with the topics coded under each theme. Some topics appeared so infrequently that they could not be grouped with others and will appear in the results despite not being included in Table I. Examples of these topics include the Bohr-Sommerfeld model and quantum Hall effect. For one institution, which had two core quantum courses, the syllabi did not offer information about the topics covered. Therefore, the topics covered in the core quantum courses were analyzed for 67 institutions, across 119 unique courses.

C. QFT course data collection and coding

To characterize QFT courses, each of the course catalogs from the 184 institutions was used to record the QFT courses offered to graduate students. In order for a course to be considered a QFT course, its title must have contained one of the following: “quantum field theory,” “field theory,” or “relativistic quantum mechanics.” As will be demonstrated in the results, the number of QFT courses offered in the course catalogs will range from zero to five courses at each institution. We attempted to obtain a syllabus from each of the recorded QFT courses, either by contacting a recent instructor of the course or by contacting administrative faculty or staff in the department. Through these communications, we obtained 91 syllabi.

The 91 syllabi were collected from a total of 52 institutions. From these 52 institutions, 40% of physics Ph.D. degrees and 13% of exiting master’s degrees were awarded in the 2023–2024 academic year [28]. The populations of these programs comprise 31% of first-year physics graduate students and 35% of the total U.S. physics graduate enrollment. These institutions are 17% MSIs, with HSI ($n = 5$), AANAPISI ($n = 4$), and AANH ($n = 1$) represented [27]. These 52 institutions are primarily public (75%) and have very high research activity (96%) [29]. The remaining institutions are classified as having high-research activity (4%).

The 91 syllabi were analyzed to determine the required textbooks for the courses. To consolidate the results, unique editions of the textbooks were not separated within the data presented in this paper.

To determine the topics taught within the QFT courses, an emergent coding method was used [30]. All topics mentioned in the syllabi were recorded. Using the table of contents from Peskin and Schroeder’s *An Introduction to Quantum Field Theory* [32] (the most commonly used textbook, as shown in the results), closely related topics were combined. Table II below presents the overarching

themes within the syllabi, along with the topics coded under each theme. Several topics did not clearly fall into these overarching themes and will therefore appear in the results despite not appearing in Table II. Due to some syllabi not containing a list of topics or course schedule, only 85 of the QFT courses were utilized for determining the course content. These 85 courses were offered across 51 different institutions.

D. Statistical analysis

To examine the relationships between institutional characteristics and graduate quantum curriculum structure, we conducted a series of quantitative analyses on the dataset of 184 institutions. All analyses were performed in R (version 4.5.3), using the tidyverse suite of packages for data manipulation and analysis. The two primary outcome variables were the number of required core quantum mechanics courses and the number of QFT courses offered.

Predictor variables included three continuous measures of program size: the number of Ph.D. degrees awarded in the 2023–2024 academic year, first-year graduate enrollment, and total graduate enrollment. Categorical predictors included public versus private institutional status, MSI versus non-MSI designation, and research activity classification (R1, R2, or other).

Pearson and Spearman correlation coefficients were computed to assess linear and monotonic relationships between Ph.D. degrees awarded and first-year graduate enrollment and each of the two curriculum outcomes (number of core quantum courses required and QFT courses offered).

To evaluate differences in curriculum structure across institutional categories, independent samples Welch’s t tests were used for the two binary categorical variables (public versus private and MSI versus non-MSI). One-way analyses of variance (ANOVAs) were used to compare the two curriculum outcomes across research activity classification (R1, R2, or other). When ANOVA results were statistically significant, Tukey’s honestly significant difference tests were used for pairwise comparisons.

To examine the combined associations between institutional characteristics and curriculum structure, separate multiple linear regression models were estimated for each curriculum outcome. Both models included the same seven predictor degrees of freedom: Ph.D. degrees awarded, first-year graduate enrollment, total graduate enrollment, public versus private status, MSI designation, and research activity classification. Research activity classification was represented by two indicator variables, with R1 institutions serving as the reference group.

Assumptions of parametric tests were evaluated using diagnostic residual plots to assess linearity, homogeneity of variance, and normality. Statistical significance was evaluated at the $\alpha = 0.05$ level. Effect sizes were reported

TABLE I. Overarching themes and their associated topics generated from emergent coding of 119 core quantum course syllabi across 67 institutions.

Kinematics Fundamentals; states; uncertainty; operators; measurement; stationary distributions; state vectors; observables; postulates; linear algebra; Hilbert space; Dirac notation. Angular momentum Angular momentum eigenstates and eigenvalues; orbital angular momentum; spherically symmetrical potentials; isotropic harmonic oscillator; addition of angular momentum; rotation; spin; Clebsch-Gordan coefficients; Pauli-two component formalism; ladder operators; spherical harmonics; Bell's inequality; Wigner-Eckart theorem; Rabi oscillations. Approximations Perturbation theory (nondegenerate, degenerate, time independent, and time dependent); WKB; adiabatic; Berry's phase; fine structure; Zeeman effect; Stark effect; variational methods; interaction picture of Hamiltonian with extreme time dependence; applications to interaction with radiation; energy shift; decay widths; Dyson series; sudden approximations; effective Hamiltonian; Fermi's golden rule; Landau-Zener formula; Lamb shift. Identical particles Permutation symmetry; symmetrization postulate; two-electron system; helium atom; multiparticle states; multiparticle wave function; fermions; bosons; indistinguishable particle; density functional theory; two particle scattering cross section. Ensembles Density matrices; many particle systems; multiparticle systems; mixed states; Bloch sphere; decoherence; entanglement; purification; reduced density matrix; two level systems; composite systems; coherence; correlation. Potentials Harmonic oscillator; wells; spherical; linear; delta; double well; periodic; tunneling. Quantum information Bit versus qubit; quantum information theory; computation; Deutsch algorithm; Grover algorithm. Quantum optics Hong-Ou-Mandel effect; Dicke model; quantum theory of light. Review of classical mechanics Galilean transformation; Hamiltonian mechanics; Lagrangian mechanics; classical mechanics relation to quantum mechanics; classical optics; Lorentz transformations in special relativity.	Dynamics State evolution; Heisenberg picture; Heisenberg equation of motion; Schrödinger picture; stationary states; Schrödinger equation; spin precession; Ehrenfest theorem; time evolution; path integrals. Symmetry Discrete; continuous; translational invariance; lattice translations; rotational invariance; space inversion; time reversal; rotation group; parity; conservation; Kramers' theorem. Scattering Scattering as time dependent perturbation theory; scattering amplitude; phase shift; Born approximation; partial waves; Eikonal approximation; low energy scattering; resonance scattering; symmetry in scattering; inelastic electron-atom scattering; Lippmann-Schwinger equation; Breit-Wigner distribution; Rutherford scattering; Ramsauer-Townsend effect. QFT Relativistic quantum mechanics; relativistic wave equation; paths to relativistic quantum mechanics; relativistic time dependent Schrödinger equation; Klein-Gordon equation; Klein-Gordon fields; Dirac equation; Dirac fields; quantization of electromagnetic fields; negative energies; Landau theory; gauge theory; Feynman diagrams; quantum electrodynamics; gauge transformations. Statistical mechanics Entropy; stochastic processes; chaos; von Neumann; entropy; Bose-Einstein; Fermi-Dirac; Gibbs distribution; Rényi entropies; Berry-Tabor conjecture; Fermi gas. Molecular Molecules; hydrogen molecules; molecular physics; Born-Oppenheimer approximation. Tunneling Tunneling through potential barrier; tunneling of particle through quasibound state; transmission; reflection. Many-body theory Green's function and perturbation theory; Thomas-Fermi model; Hartree-Fock framework. Experiments Stern-Gerlach; photoelectric effect; blackbody radiation; Davisson-Germer; double slit.
---	--

TABLE II. Overarching themes and their associated topics generated from emergent coding of 85 QFT syllabi across 51 institutions.

Fundamentals Lorentz transformations; Lagrangian mechanics; Hamiltonian mechanics; Green's functions; two-point function; N-point functions; correlation function; propagators; complex integration; pole prescriptions; groups; functional derivatives; generating functionals; spectral decomposition; field integrals. Dirac field Lorentz invariance; Dirac equation; Weyl spinors; spin sums; Dirac matrices; gamma matrices; Dirac field bilinear; Dirac propagator; quantization of Dirac field; Dirac theory; spin; chiral representations. Quantum electrodynamics Trace technology; unpolarized cross section; crossing symmetry; Mandelstam variables; Compton scattering; photon polarization sums; Klein-Nishina formula; Møller scattering; Bhabha scattering; photon field; gauge field; minimal coupling; dielectric function; Maxwell's equations. Functional methods Path integrals; functional quantization of scalar fields and spinor fields; functional formalism symmetry; functional integrals; photon propagator; Schwinger-Dyson equation; partition function; Grassmann numbers; Grassmann variables. Non-Abelian gauge theories Yang-Mills Lagrangian; gauge invariance; Wilson loop; Lie algebras; Casimir operator; confinement; non-Abelian gauge bosons; Faddeev-Popov Lagrangian; BRST symmetry; gauge boson self-energy; asymptotic freedom; ghosts. Topology Topological defects; topological QFTs; Chern-Simons; topology of electronic bands; topological insulators; topological bosonic excitations; topological transitions. Perturbation theory anomalies Axial current in 2D/4D; vacuum polarization diagrams; current operator equation; triangle diagrams; Goldstone bosons; chiral symmetries; chiral anomalies; chiral gauge theories. Lattice field theory Discrete Fourier transforms; deterministic lattice field theory.	Klein-Gordon field Classical field theory; Lagrangian field theory; Hamiltonian field theory; Noether's theorem; Noether's current; Klein-Gordon equation; Poincaré group. Interacting fields and Feynman diagrams Perturbation theory; perturbation expansion of correlation function; Wick's theorem; cross sections; S-matrix; Feynman rules; Yukawa theory; Coulomb potential; ϕ^4 theory; scalar field theory; vacuum bubbles; connected diagrams; linked-cluster theorem; Born approximation; decay rates. Radiative corrections Soft bremsstrahlung electron vertex; Feynman parameters; electron self-energy; LSZ reduction; optical theorem; Ward-Takahashi identity; Wick rotation; running couplings; dimensional regularization; one loop diagrams; tree diagrams; tree-level processes; vertex corrections; Källén-Lehmann form; infrared divergences; Cutkosky rules; Lamb shift; anomalous magnetic moment; Pauli-Villars regularization. Renormalization Renormalization group; ultraviolet divergences; renormalized perturbation theory; superficial degree of divergence; linear sigma model; Goldstone's theorem; effective action; phase transitions; Legendre transforms; solitons; Wilson's approach; Callan-Symanzik equation; coupling constants; evolution of mass parameters; effective mass; screening. Quantum chromodynamics Deep inelastic scattering; distribution functions; hard-scattering processes in hadron collisions; lepton pair production; jet pair production; parton evolution; Altarelli-Parisi equations; Gell-Mann and Low theorem; strong coupling. String theory Holographic dualities; AdS/CFT. Gauge theory spontaneous symmetry breaking Higgs mechanism; Glashow-Weinberg-Salam theory; Higgs boson; quark and lepton symmetry; electroweak theory; Goldstone boson equivalence.
---	--

according to the corresponding analysis: Pearson's r and Spearman's ρ for correlation analyses, R^2 and adjusted R^2 for regression models, Cohen's d for binary group comparisons and eta-squared (η^2) for one-way ANOVAs.

IV. RESULTS

A. Core quantum courses

Based on the course catalogs, graduate bulletins, and/or physics department graduate handbooks from the 184 institutions, the number of core quantum courses is shown

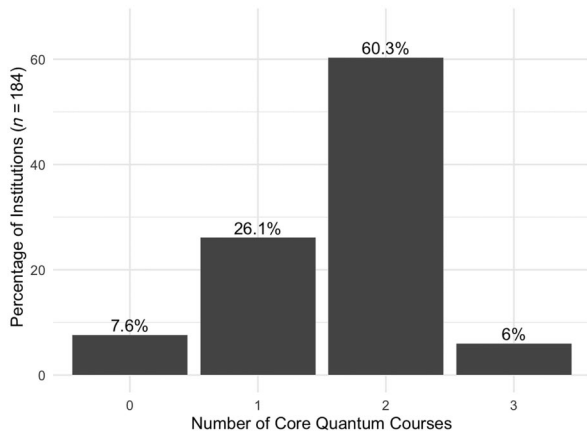


FIG. 1. Number of required core quantum courses at 184 graduate physics programs.

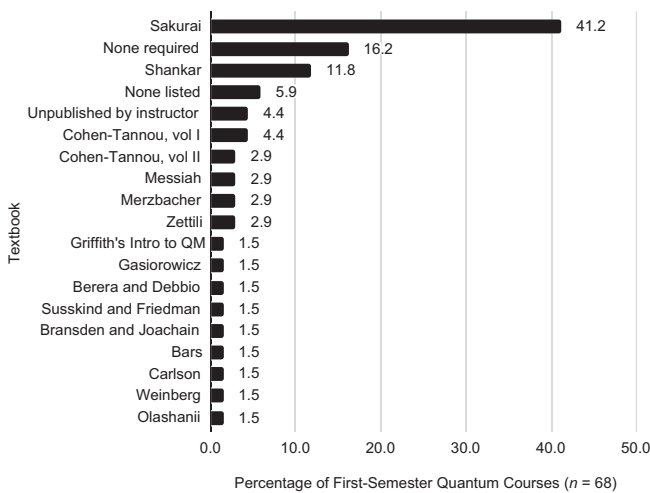


FIG. 2. Textbook required for first-semester graduate quantum mechanics courses across 68 institutions.

in Fig. 1. The majority of graduate physics programs in this study required two core quantum courses (60%). One core quantum course was required by 26% of institutions, while 7% did not require any core quantum courses, and 6% required three.

The textbook analysis (Fig. 2) revealed that, across the 68 institutions requiring one or more core quantum courses, Sakurai's *Modern Quantum Mechanics* was the most commonly assigned textbook for the first-semester core course, used at 41.2% of institutions. The next most common occurrence was to not require or not specify a textbook for the course at all (16.2%). An interesting occurrence that was not found in a similar undergraduate-level quantum mechanics textbook analysis [9] was that instructors would utilize a textbook they were working toward publishing themselves (4.4%).

Among the 67 institutions analyzed for themes and topics in the core quantum curriculum, the most commonly included topics in course syllabi were kinematics (99% of curricula), angular momentum (91% of curricula), approximations (91% of curricula), and dynamics (90% of curricula). This pattern is consistent, as these topics correspond to the first four chapters of Sakurai [31]. The results of the themes analysis can be seen in Fig. 3. However, there were 69 topics that were only introduced within one institution's curricula and therefore would only account for 1.25% of institutions. These topics can be found in Table III.

B. QFT courses

From our analysis of the course catalogs of the 184 institutions, it was found that 71 institutions offered zero QFT courses, 35 institutions offered one course, 50 institutions offered two courses, 23 institutions offered three courses, 4 institutions offered four courses, and one institution offered 5 courses.

From the 91 QFT course syllabi, the most commonly (30%) required textbook was by Peskin and Schroeder [32]. The other required textbooks can be found in Fig. 4. Due to some courses requiring more than one textbook, the total percentage in the graph adds to more than 100%.

From the QFT syllabi that themes could be discerned from, the most commonly included themes were interacting fields and Feynman diagrams (72%), quantum electrodynamics (QED) (71%), and radiative corrections (68%). The complete results are shown in Fig. 5. There were 34 topics that appeared in only one syllabus and did not have a direct connection to belong as a code in the overarching themes. These 34 topics are displayed in Table IV.

C. Statistical analysis

Pearson and Spearman correlations were used to examine relationships between the continuous measures of program size (Ph.D. degrees awarded and first-year enrollment) and the number of core quantum courses. No significant relationship was observed between the number of core quantum courses and either Ph.D. degrees awarded (Pearson's $r = 0.034$, $p = 0.664$; Spearman's $\rho = 0.109$, $p = 0.160$) or first-year graduate enrollment (Pearson's $r = -0.018$, $p = 0.819$; Spearman's $\rho = 0.040$, $p = 0.602$).

In contrast, the number of QFT courses showed a moderate positive relationship with both Ph.D. degrees awarded (Pearson's $r = 0.478$, $p < 0.001$; Spearman's $\rho = 0.514$, $p < 0.001$) and first-year enrollment (Pearson's $r = 0.364$, $p < 0.001$; Spearman's $\rho = 0.424$, $p < 0.001$), indicating that larger programs tend to offer more advanced quantum coursework.

Welch's two-sample t tests were conducted to compare the binary institutional variables (public versus private and MSI versus non-MSI) with curriculum structure. No statistically significant differences were observed in the

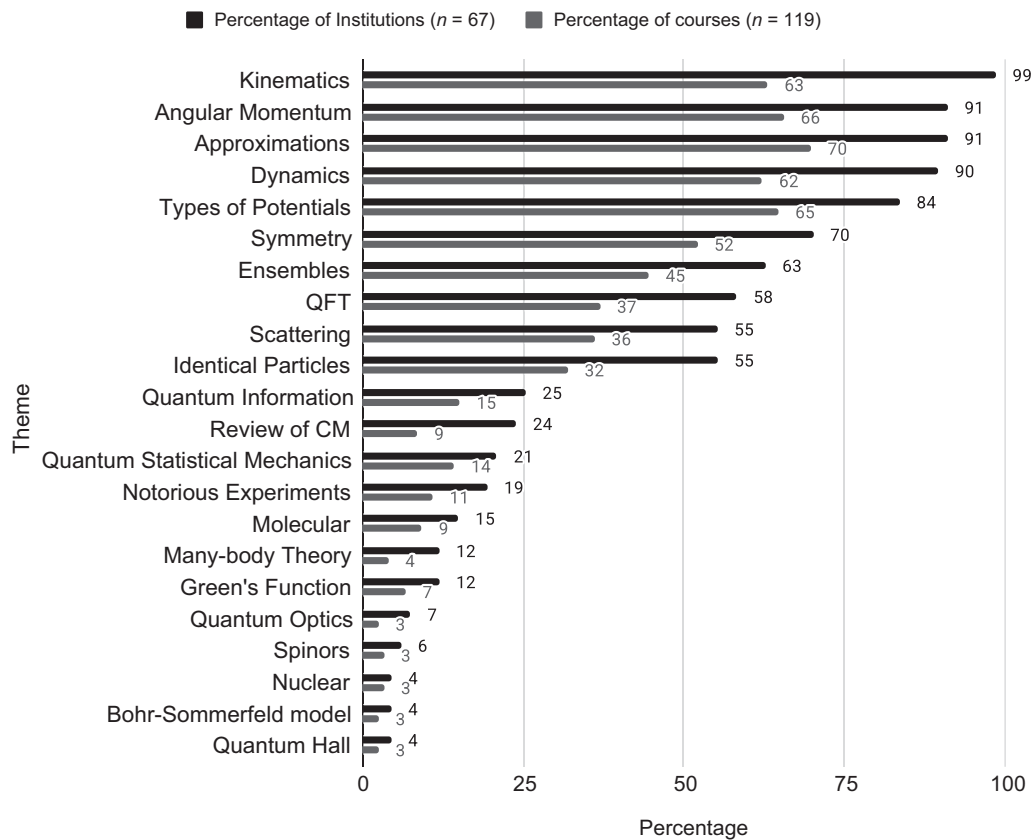


FIG. 3. Themes identified in the core quantum curricula of 67 institutions. Dark gray represents the percentage of institutions (67 total), with courses treated in aggregate at the institutional level. Light gray represents the percentage of courses (119 total), with individual courses treated as separate entries.

number of core quantum courses between public and private institutions (public: $M = 1.66$, private: $M = 1.61$, $p = 0.609$), nor between MSI and non-MSI institutions (MSI: $M = 1.76$, non-MSI: $M = 1.62$, $p = 0.369$).

Similarly, the number of QFT courses did not differ significantly between public and private institutions (public: $M = 1.15$, private: $M = 1.38$, $p = 0.239$), or between MSI and non-MSI institutions (MSI: $M = 1.30$, non-MSI: $M = 1.20$, $p = 0.678$). Together, these results suggest that institutional classification is not strongly associated with differences in quantum course requirements or QFT offerings across programs.

One-way ANOVAs were used to compare institutions across research activity classification (R1, R2, or other). For core quantum courses, a statistically significant effect of research activity was observed, $F(2, 181) = 3.55$, $p = 0.031$. Tukey *post hoc* comparisons indicated that institutions classified as “high research activity” (commonly referred to as R2) required fewer core quantum courses than institutions classified as “very high research activity” (commonly referred to as R1) (mean difference = -0.34 , $p = 0.023$), while no other pairwise comparisons reached statistical significance.

For QFT courses, a stronger effect of research activity was observed, $F(2, 181) = 10.64$, $p < 0.001$. *Post hoc* analyses revealed that R2 institutions offered fewer QFT courses than R1 institutions (mean difference = -0.86 , $p < 0.001$).

Multiple linear regression models were estimated to examine the associations between institutional characteristics and each curriculum outcome. Both models included Ph.D. degrees awarded, first-year enrollment, total graduate enrollment, public versus private status, MSI designation, and research activity classifications as predictors.

The model for core quantum courses was not statistically significant, $F(7, 158) = 1.19$, $p = 0.312$, and explained little variance in the outcome (adjusted $R^2 = 0.008$). The only significant predictor was research activity classification: R2 institutions were associated with fewer core quantum courses relative to R1 institutions ($\beta = -0.40$, $p = 0.010$). Ph.D. degrees awarded, first-year enrollment, total graduate enrollment, public and private status, and MSI designation were not statistically significant predictors.

In contrast, the model predicting QFT course offerings was statistically significant, $F(7, 158) = 10.61$, $p < 0.001$,

TABLE III. Topics introduced at only one institution within the core quantum courses (1.25% of institutions).

Topics introduced by only one institution
Adiabatic gauge potential
Antiferromagnetism
Asymptotic analysis
BCS theory
Berry conjecture
Berry curvature
Bohigas-Giannoni-Schmit conjecture
Bopp representation
Chern number
Cyclotron resonance
Dirac equation symmetries
Discrete nonlinear Schrödinger equation
EBK approximation
Fermi-Hubbard model
Foldy-Wouthuysen transformation
Fubini-Study metric
Hanbury Brown and Twiss effect
Hellmann-Feynman theorem
Hole theory
Hubbard model
Husimi Q representation
Ion traps
Jaynes-Cummings model
Jordan-Wigner transformation
Jump operators
Kubo formula
Landauer theory
Laplace-Runge-Lenz vector
Lieb-Liniger model
LIGO
Lindblad equation
Local realism
Lorentz group
Many worlds
Masers
Monochromatic perturbations
Moyal product
Nagaoka ferromagnetism
Neutrinos
Noether's theorem
Number-phase representation
Objective reality
Occupation number
Phi-4/quartic interaction
Phonons
Photoproduction of hydrogen
Quantum gas microscope
Quantum fields
Quantum Fisher information
Random matrix theory
Renormalization group
Resolvent operator
Schur's lemma
Spontaneous symmetry breaking

(Table continued)

TABLE III. (Continued)

Topics introduced by only one institution
Standard model
String theory
Superconducting quantum interference devices
Superfluidity
Superoperators
Theory of continuous groups
Time-ordered product
Tonks-Girardeau gas
Two-site Bose-Hubbard model
Wigner-Brillouin perturbation theory
Wigner surmise
Wigner-Weyl transform
Young tableau
Yukawa potential
Zeno effect

and explained a substantial proportion of variance (adjusted $R^2 = 0.290$). Significant predictors included first-year enrollment ($\beta = -0.035$, $p = 0.012$), total graduate enrollment ($\beta = 0.012$, $p < 0.001$), and research classification, with R2 institutions offering fewer QFT courses than R1 institutions ($\beta = -0.54$, $p = 0.014$). Ph.D. degrees awarded, public and private status, and MSI designation were not statistically significant predictors. These results suggest that QFT curriculum structure is more strongly associated with institutional size and research classification than core quantum coursework.

Effect size estimates further clarified the magnitude of observed relationships. For the binary institutional characteristics, Cohen's d (-0.19 to -0.08) indicated small differences in course offerings across public and private and MSI and non-MSI institutions. For research activity classification, η -squared indicated a small effect for core quantum course requirements ($\eta^2 = 0.04$) and a moderate effect for QFT course offerings ($\eta^2 = 0.11$). In contrast, QFT coursework demonstrated small-to-moderate effect sizes ($\eta^2 = 0.11$) and a moderate correlation with Ph.D. production ($\rho = 0.51$), indicating substantively meaningful variation across institutions.

Together with regression results (adjusted $R^2 = 0.29$ for QFT versus 0.008 for core quantum courses), these findings suggest that core quantum courses are relatively standardized across institutions, whereas QFT curricula exhibit structural variation associated with institutional characteristics.

V. DISCUSSION

This study provides the first comprehensive national map of graduate-level quantum instruction in U.S. physics doctoral programs. Our findings revealed the core quantum

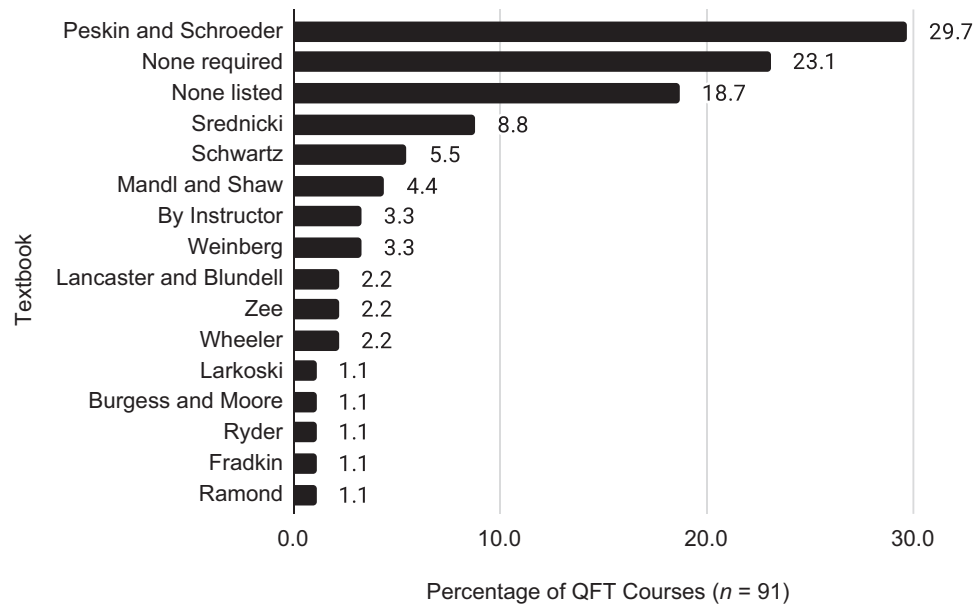


FIG. 4. Textbooks required for 91 QFT courses offered at 52 institutions. The total percentage within the graph adds to more than 100%, due to some courses requiring more than one text.

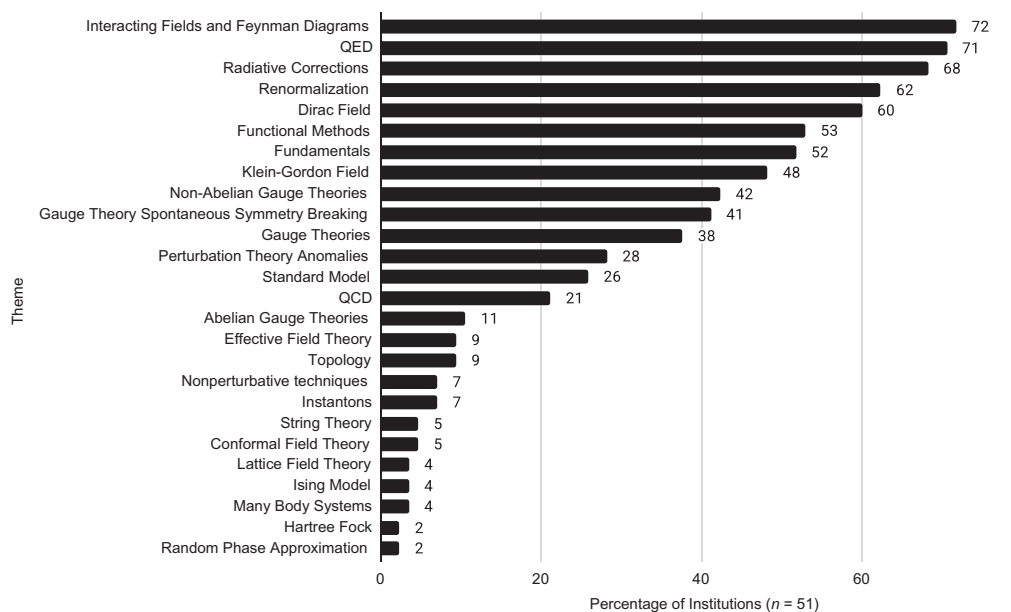


FIG. 5. Themes present in the QFT curricula of 51 different institutions.

curriculum is anchored by a canonical set of four topics: kinematics (99% of curricula), angular momentum (91% of curricula), approximations (91% of curricula), and dynamics (90% of curricula). These topics are most often taught through Sakurai's *Modern Quantum Mechanics* [31]. From Fig. 3, these canonical topics form a plateau of the most commonly taught content and therefore likely represent material that is commonly introduced in the first semester of a typical graduate quantum sequence. A second plateau may correspond to topics more commonly

introduced in a second-semester course, including symmetry (70%), ensembles (63%), QFT (58%), scattering (55%), and identical particles (55%). In this context, the inclusion of QFT refers to the presence of QFT concepts within the graduate core quantum curriculum rather than the availability of a dedicated QFT course, suggesting that many programs provide only introductory exposure to this material within the core sequence.

Figure 3 also illustrates the difference between the percentage of institutions in which each topic appears

TABLE IV. Topics only explicitly mentioned in one of the QFT syllabi.

Topics found in only one QFT syllabus
Quantum Hall effect
Spin waves
Interacting electron gas
Collective excitations
Zero modes
Cabibbo-Kobayashi-Maskawa matrix
Lepton and baryon number violation
Nonperturbative lattice formulation
Blackbody
Froissart bound
Inhomogeneous vacuum field configurations
Gervais-Neveu gauge
Random walks
Mean field theory
Aharonov-Bohm effect
Wess-Zumino-Novikov-Witten models
Dark matter
Cosmic inflation
Quantum gravity and black holes
Bogomol'nyi completion
Matsubara representation
Keldysh-Schwinger formalism
Ballistic versus disordered dynamics
Random matrix theory
Wigner-Dyson versus Poisson level statistics
Quantum dots
Complex nuclei
Spin glasses
Sherrington-Kirkpatrick
Sachdev-Ye-Kitaev
Theory of localization
Finite- N techniques
Glashow-Iliopoulos-Maiani mechanism
Grand unification

(67 institutions total, in dark gray) and the percentage of individual courses in which each topic appears (119 courses total, in light gray). Because the course-level analysis treats all courses in a program's graduate core sequence equally, the normalization includes both introductory and later courses. Consequently, foundational topics that are primarily taught in the first course of a multicourse sequence will naturally appear in a smaller fraction of courses than topics that are revisited throughout the sequence. The course-level percentages should therefore be interpreted as reflecting how broadly a topic is distributed across the graduate core curriculum rather than the likelihood that it is taught in an introductory course.

While kinematics was the most commonly represented topic at the institutional level, approximations were the most commonly represented topic when individual courses were disaggregated. In a subsequent study, faculty reported that approximations were an important concept for

research and that they often needed to supplement formal instruction on this topic when mentoring graduate researchers [26]. Taken together, these two findings suggest that approximations should be explored in greater depth within the core graduate quantum curricula.

The emergence of a canonical set of core quantum topics provides insight into the implicit expectations graduate programs hold about incoming student preparation. However, prior research [9] has documented substantial variability in undergraduate quantum curricula across the same institutions analyzed in this study. Prior work concluded the average time spent on quantum topics in the four-year undergraduate curricula to be 63.5 h, but with a large standard deviation of 28.1 h. While almost all curricula explicitly included the Schrödinger equation (98%), only 46% included perturbation theory (an approximation method). These prior results demonstrate the variability in depth and rigor of undergraduate quantum curricula. As a result, while graduate quantum curricula may appear to build directly on undergraduate content, they may not consistently meet students where they are, instead presuming a level of conceptual and mathematical preparedness that is unevenly distributed among incoming students. While it may have previously been assumed that students at the graduate-level are prepared and can be expected to engage with abstract, formal treatments of familiar concepts, the topics we had assumed to be “familiar” may not be so familiar after all.

Sakurai's [31] treatment of Dirac notation provides another mismatch between undergraduate and graduate studies to highlight. While it was found that 34% of undergraduate institutions teach Dirac notation [9], Sakurai starts from an assumed fluency in this notation [31]. Sakurai introduces concepts and provides definitions utilizing arbitrary states and generalized equations. Sakurai does not provide examples of these equations in context. While it may have previously been assumed that graduate students were prepared for abstract and generalized concepts as a result of working with more concrete examples in their undergraduate education, we now know that some graduate students may not have seen these concepts previously. Previous research has demonstrated that even after graduate-level instruction in quantum mechanics, many physics students have difficulties applying Dirac notation [33]. But perhaps, this is because graduate-level instruction is some students' first encounter with Dirac notation. This is particularly important because Dirac notation is fundamental to how expert physicists do quantum mechanics and important for a rising physicist to learn. We recommend that instructors and programs integrate Dirac notation at the undergraduate-level, and that graduate coursework provides an introduction to Dirac notation that meets a variety of student backgrounds. If an instructor chooses to utilize Sakurai [31], we recommend they introduce contextualized examples of the generalized equations to their students during instruction.

Faculty expectations in graduate quantum courses and assessments may presume prior familiarity, meaning students may struggle not from lack of ability but from differences in preparation. This misalignment may help explain why some students report stress and even trauma around qualifying exams [34], which most commonly test students on their quantum knowledge [7]. These findings therefore highlight the need for greater alignment between undergraduate preparation and graduate expectations, or for graduate programs to more explicitly account for variation in students' prior exposure.

While characterizing the core quantum curricula in graduate programs was important, it is just the first step. Future work will characterize other important courses within physics programs, such as electricity and magnetism, classical mechanics, and statistical mechanics. The goal would be to capture all of the core courses within physics to better understand the training physicists receive.

Core quantum courses showed no statistically significant differences based on institution size or type, with the exception that R2 institutions required fewer courses than R1 institutions. In contrast, QFT, as an advanced quantum course, exhibited a different pattern. QFT was not offered in the course catalogs of more than one-third of the institutions analyzed. R2 and smaller graduate programs were more likely to lack any QFT offerings compared to R1 and larger programs. This pattern raises questions about equitable access to QFT training across institutions with different enrollment levels. Students enrolled in programs without a formal QFT course must therefore rely on informal pathways for learning and preparation.

For institutions that did offer QFT courses, a canonical set of topics emerged: interacting fields, Feynman diagrams, QED, and radiative corrections. This canon offers a structured basis for institutions seeking to broaden their advanced quantum offerings through the inclusion of a dedicated QFT course.

VI. LIMITATIONS

Syllabi were used for this study because they are either publicly available or readily obtainable and they impose a relatively low time burden on instructors who choose to participate. However, there is a possible selection effect in our result based on the instructors willing to share their syllabi with us when not publicly available.

Syllabi were chosen for this study because they can reflect instructors' epistemological perspectives and pedagogical approaches [35], thereby providing insight into curricular decisions and institutional norms. However, syllabi should be interpreted as approximations of enacted courses, as the pace of instruction may vary and influence which topics are ultimately covered. In this sense, syllabi may be either "overambitious" or "conservative" in their representation of course content. Additionally, syllabi feature variability from institution-to-institution and even

instructor-to-instructor within the same institution. Some instructors may not provide a list of topics covered, thereby narrowing the available syllabi for use in the topic-characterization section of this study (as mentioned in the Methods). The institutional sample decreased from 184 institutions, representing 84% of enrolled physics Ph.D. students in the U.S. at the course catalog level, to 67 institutions, representing 34% of enrolled physics Ph.D. students at the syllabus topic-characterization level.

A further limitation is that syllabus analysis relies on explicitly stated content, which may not fully capture what is actually taught. For example, in this study, kinematics and dynamics were identified in 99% and 90% of curricula, respectively. These values are likely underestimates, as it is difficult to imagine a graduate quantum course that does not include foundational material associated with kinematics or discussions of time evolution associated with dynamics. This discrepancy raises epistemological questions about the distinction between what instructors choose to make explicit and what they assume to be implicit in course design.

VII. CONCLUSION

This study offers the first national-scale map of graduate-level quantum mechanics and QFT instruction across 184 U.S. doctoral-granting physics programs. We mapped both the shared foundations and variability that shape graduate quantum education. Core quantum mechanics courses show strong convergence around a canonical set of topics and a common textbook, whereas QFT instruction is unevenly available. These findings provide a foundation for rethinking how graduate coursework supports students' transitions into advanced study and research. They also highlight structural features that may have implications for preparation and opportunity, particularly for students whose undergraduate experiences did not emphasize key formalisms such as Dirac notation or who enter programs without access to formal QFT instruction.

Characterizing the graduate quantum curriculum was the first step in this investigation. Future research should examine the content of other graduate courses, as well as the motivations behind curricular and textbook choices, the experiences of students navigating these structures, and the role of departmental culture in shaping educational pathways. Understanding these dynamics will be essential for ensuring that graduate quantum education remains rigorous, equitable, and aligned with the needs of an increasingly quantum-enabled scientific and technological landscape.

DATA AVAILABILITY

The data that support the findings of this article are not publicly available because they contain sensitive personal information. The data are available from the authors upon reasonable request.

- [1] M. Verostek, C. W. Miller, and B. Zwickl, Analyzing admissions metrics as predictors of graduate GPA and whether graduate GPA mediates PH.D. completion, *Phys. Rev. Phys. Educ. Res.* **17**, 020115 (2021).
- [2] N. T. Young, K. Tollefson, R. G. T. Zegers, and M. D. Caballero, Rubric-based holistic review: A promising route to equitable graduate admissions in physics, *Phys. Rev. Phys. Educ. Res.* **18**, 020140 (2022).
- [3] N. T. Young, N. Verboncoeur, D. C. Lam, and M. D. Caballero, Rubric-based holistic review represents a change from traditional graduate admissions approaches in physics, *Phys. Rev. Phys. Educ. Res.* **19**, 010134 (2023).
- [4] N. J. Mikkelsen, N. T. Young, and M. D. Caballero, Investigating institutional influence on graduate program admissions by modeling physics graduate record examination cutoff scores, *Phys. Rev. Phys. Educ. Res.* **17**, 010109 (2021).
- [5] C. Miller and K. Stassun, A test that fails, *Nature (London)* **510**, 303 (2014).
- [6] B. Bridges, C. Solis, J. Barron, J. Lavery, J. Chini, and R. Henderson, Investigating the assessment landscape of physics graduate programs, in *Physics Education Research Conference 2023*, PER Conference (Sacramento, CA, 2023), pp. 64–69, [10.1119/perc.2023.pr.Bridges](https://doi.org/10.1119/perc.2023.pr.Bridges).
- [7] D. Campbell, T. Appelquist, R. Diehl, J. Fajans, J. D. Garcia, J. Gates, A. Goldman, P. Jung, and M. Paesler, Report of the joint AAPT-APS task force: Graduate education in physics, American Association of Physics Teachers and American Physical Society, College Park, MD, Technical Report, 2006.
- [8] A. Buzzell, T. J. Atherton, and Rón Barthelemy, Modern physics courses: Understanding the content taught in the U.S., *Phys. Rev. Phys. Educ. Res.* **21**, 010139 (2025).
- [9] A. Buzzell, T. J. Atherton, and Rón Barthelemy, Quantum mechanics curriculum in the U.S.: Quantifying the instructional time, content taught, and paradigms used, *Phys. Rev. Phys. Educ. Res.* **21**, 010102 (2025).
- [10] G. Zhu and C. Singh, Students' understanding of stern gerlach experiment, *AIP Conf. Proc.* **1179**, 309 (2009).
- [11] G. Passante, Energy measurement resources in spins-first and position-first quantum mechanics, in *Physics Education Research Conference 2016*, PER Conference (Sacramento, CA, 2016), p. 236, [10.1119/perc.2016.pr.054](https://doi.org/10.1119/perc.2016.pr.054).
- [12] H. R. Sadaghiani and J. Munteanu, Spin first instructional approach to teaching quantum mechanics in sophomore level modern physics courses, in *Physics Education Research Conference 2015*, PER Conference (College Park, MD, 2015), p. 287, [10.1119/perc.2015.pr.067](https://doi.org/10.1119/perc.2015.pr.067).
- [13] J. Kruse, M. Griston, and B. R. Wilcox, Characterization of upper-level undergraduate quantum mechanics courses in the U.S., *Phys. Rev. Phys. Educ. Res.* **21**, 020162 (2025).
- [14] D. J. Griffiths, *Introduction to Quantum Mechanics*, 3rd ed. (Cambridge University Press, Cambridge, UK, 2018).
- [15] D. H. McIntyre, *Quantum Mechanics: A Paradigms Approach* (Pearson, Boston, MA, 2012).
- [16] J. Townsend, *A Modern Approach to Quantum Mechanics* (University Science Books, Sausalito, CA, 2000).
- [17] J. C. Meyer, G. Passante, S. J. Pollock, and B. R. Wilcox, Today's interdisciplinary quantum information classroom: Themes from a survey of quantum information science instructors, *Phys. Rev. Phys. Educ. Res.* **18**, 010150 (2022).
- [18] A. Piña, M. Verostek, B. Boyle, E. Watts, M. Lawler, M. Cacheiro, N. Pradeep, C. West, S. El-Adawy, H. Lewandowski, B. Zwickl, United States quantum education landscape dataset and visualizations (2024), <https://quantumlandscape.streamlit.app/>.
- [19] J. C. Meyer, G. Passante, S. J. Pollock, B. R. Wilcox1, Introductory quantum information science coursework at US institutions: Content coverage, *EPJ Quantum Technol.* **11**, 16 (2024).
- [20] D. Tong, *Lectures on quantum field theory Part III Mathematical Tripos* (University of Cambridge, 2006), <https://www.damtp.cam.ac.uk/user/tong/qft.htm>.
- [21] A. Sirnoorkar, A. Mazumdar, and A. Kumar, Towards a content-based epistemic measure in physics, *Phys. Rev. Phys. Educ. Res.* **16**, 010103 (2020).
- [22] M. Verostek, C. W. Miller, and B. M. Zwickl, Modeling when and how physics Ph.D. students search for a research group: The role of interests and prior research experiences in timely group integration, *Phys. Rev. Phys. Educ. Res.* **20**, 020119 (2024).
- [23] G. P. Lepage, Active learning in a graduate quantum field theory course, *Am. J. Phys.* **89**, 317 (2021).
- [24] A. Buzzell, T. J. Atherton, and R. Barthelemy, Assessing a combined human coding and natural language processing method for qualitative analysis in physics education research, in *Proceedings of the Physics Education Research Conference 2025* (American Association of Physics Teachers, Washington, DC, 2025).
- [25] USNEWS, <https://www.usnews.com/best-graduate-schools/top-science-schools/physics-rankings>.
- [26] A. Buzzell, T. J. Atherton, and R. Barthelemy, "Survey of physics faculty insights on the US quantum curriculum and research need" (to be published).
- [27] MSI, <https://inclusion.uci.edu/core-programs/minority-thriving-campus/msi-directory/>.
- [28] S. Nicholson and P. Mulvey, Roster of physics departments with enrollment and degree data, <https://www.aip.org/statistics/roster-of-physics-departments-with-enrollment-and-degree-data-2024>.
- [29] <https://carnegieclassifications.acenet.edu/>.
- [30] J. Creswell, *Qualitative Inquiry and Research Design: Choosing Among Five Approaches* (Sage Publications, Thousand Oaks, CA, 1998).
- [31] J. J. Skurai and S. F. Tuan, *Modern Quantum Mechanics* (Benjamin/Cummings Pub., Menlo Park, CA, 1985).
- [32] M. E. Peskin and D. V. Schroeder, *An Introduction to Quantum Field Theory* (Addison-Wesley Publishing Company, Reading, MA, 1995).
- [33] C. Singh and E. Marshman, Investigating student difficulties with dirac notation, in *Physics Education Research Conference 2013*, PER Conference (Portland, OR, 2013), pp. 345–348, [10.1119/perc.2013.pr.074](https://doi.org/10.1119/perc.2013.pr.074).
- [34] S. Basir and E. Burkholder, Departmental case study of physics doctoral students' perspectives of written qualifying exams, *Phys. Rev. Phys. Educ. Res.* **20**, 020123 (2024).
- [35] L. Parson, Are stem syllabi gendered? A feminist critical discourse analysis, *Qualitative Rep.* **21**, 102 (2016).